

Problem Session III: Discussion

Warm-up question

What is the difference between conditional and unconditional probability?

Question 1

A university surveyed students about whether they own a car and whether they live on campus.

The following information was collected:

60% of students own a car. 15% of students both own a car and live on campus. 43% of students live on campus.

Construct a probability tree diagram (see page 6 of your guided notes) based on this information.

Key talking points

We can't define any probabilities without defining events. We know:

C = student owns a car L = student lives on campus

We then also know the compliments of each!

C^c = student does not own a car

L^c = student does not live on campus.

Next, let's write out what we do know:

$$P(C) = 0.60$$

$$P(L) = .43$$

$$P(C \cap L) = .15$$

We also know by the rule of compliments that

$$P(C^c) = .4$$

$$P(L^c) = .57$$

Now let's start with event C.

Start

C (0.60)

C (0.40)

When building a tree diagram, we know that the next branch is a **conditional** probability given the event that we already calculated the probability for. So, we are interested in...

Start

C (0.60)

L | C

Lc | C

Cc (0.40)

L | Cc

Lc | Cc

By the definition of conditional probability, we can re-write $P(L | C)$ as follow!

$$P(L | C) = \frac{P(L \cap C)}{P(C)}$$

And we know each of these values!

$$= \frac{0.15}{0.60} = 0.25$$

This tells us exactly what $P(L^c|C)$ is! Based on the conditional complement rule, we know that

$$P(L^c|C) + P(L|C) = 1$$

$$P(L^c|C) + 0.25 = 1$$

$$P(L^c|C) = .75$$

Start

C (0.60)
L | C (.25)
Lc | C (.75)

Cc (0.40)
L | Cc
Lc | Cc

Now, let's calculate the $P(L|C^c)$. We are again going to use the definition of conditional probability.

$$P(L|C^c) = \frac{P(L \cap C^c)}{P(C^c)}$$

I know the $P(C^c)$! I don't know the numerator though.... But I do know some pieces that can help me calculate it. Let's explore based on the events.

We know that $L = (L \cap C) \cup (L \cap C^c)$ because the outcome of the events are disjoint! That is, a student can not own a car and not own a car at the same time!

Because of this, we can use the **law of total probability** to rewrite this into the following:

$P(L) = P(L \cap C) + P(L \cap C^c)$ and I know two of these three probabilities!

$$.43 = .15 + P(L \cap C^c)$$

$$P(L \cap C^c) = .28$$

So...

$$P(L|C^c) = \frac{.28}{.40} = .70$$

And because of conditional complements, we know that...

$$P(L^c|C^c) = \frac{.28}{.40} = .3$$

Start

C (0.60)
L | C (.25)
Lc | C (.75)

Cc (0.40)
L | Cc (.70)
Lc | Cc (.30)

Takeaway: Use what you know as a clue on how to explore/rewrite probabilities that you are trying to solve!

Question 2

A department store sells sport shirts in three sizes (small, medium, and large), three patterns (plaid, print, and stripe), and two sleeve lengths (long and short). The accompanying table gives the proportions of shirts sold in the various category combinations.

Length	Size	Plaid	Print	Striped
Short	Small	0.04	0.02	0.05
Short	Medium	0.08	0.07	0.12
Short	Large	0.03	0.07	0.08
Long	Small	0.03	0.02	0.03
Long	Medium	0.10	0.05	0.07
Long	Large	0.04	0.02	0.08

- (a) What is the probability that the next shirt sold is a medium, long-sleeved, print shirt? (2 points)

The outcomes of each of our three events are defined! Thus, we can “simply” find the probability we are looking for in the table based on the context.

Length	Size	Plaid	Print	Striped
Long	Medium	0.10	0.05	0.07

$$P(\text{medium} \cap \text{long} \cap \text{print}) = 0.05$$

- (b) What is the probability that the next shirt sold is a medium print shirt? (4 points)

First, we start out by writing the probability statement we want to solve for: $P(\text{medium} \cap \text{print})$. Next, we ask ourselves, “how many ways can we see a medium print shirt”? The answer is two, as this type of shirt can either be length short or long. **Because the outcomes of the event are disjoint**, we can use additivity: $P(\text{medium} \cap \text{print}) = P(\text{medium} \cap \text{print} \cap \text{short}) + P(\text{medium} \cap \text{print} \cap \text{long})$. Again, turns into a game of locate the correct probabilities, and add them together.

$$= 0.07 + 0.05 = .12$$

(c) What is the probability that the next shirt sold is a long sleeved shirt? (4 points)

Use the same logic in b to answer this question!

(d) What is the probability that the next shirt sold has a print pattern? (4 points)

use the same logic in b to answer this question

(e) **If** the next shirt sold was a short sleeved plaid, what is the **probability** it was of medium size? (4 points)

I've highlighted two words in this problem that I want us to focus on: “If” and “probability”. These words give us clues as to how we can build our probability statement we are trying to find.

Words like **if**, **conditioned**, **given**, help us understand an outcome of an event that has already happened. This goes to the right of the | in our conditional statement.

probability clues us into what we are actually trying to calculate the probability of, and what will go to the left of our |.

Thus, we are trying to solve: $P(\text{medium}|\text{short} \cap \text{plaid})$

And we can rewrite this using the definition of conditional probability

$$\frac{P(\text{medium} \cap \text{short} \cap \text{plaid})}{P(\text{short} \cap \text{plaid})}$$

Now we can ask ourselves, what do we know? Well I know $P(\text{medium} \cap \text{short} \cap \text{plaid})$ because each event has an outcome defined! I don't know $P(\text{short} \cap \text{plaid})$, but I have everything I need to solve it.

– I know the outcomes of **Size** are disjoint (can't be large and a small shirt)

– I know that a short plaid shirt can happen three different ways (small, medium, and large)

Thus, based on additivity...

$$\frac{P(\text{medium} \cap \text{short} \cap \text{plaid})}{P(\text{short} \cap \text{plaid} \cap \text{small}) + P(\text{short} \cap \text{plaid} \cap \text{medium}) + P(\text{short} \cap \text{plaid} \cap \text{large})}$$

And I can find these values from the table!

$$\frac{.08}{.04+.08+.03} = .533$$

- (f) If the next shirt sold was a medium plaid, what is the probability it was long sleeved? (4 points)

use the same logic as in e to answer the question

Question 3

What would make this an invalid joint probability distribution?

Try and think of two potential ideas...

Length	Size	Plaid	Print	Striped
Short	Small	0.04	0.02	0.05
Short	Medium	0.08	0.07	0.12
Short	Large	0.03	0.07	0.08
Long	Small	0.03	0.02	0.03
Long	Medium	0.10	0.05	0.07
Long	Large	0.04	0.02	0.08

Try this question on your own, and reach out if you have any questions!

Question 4

A city survey categorizes households by pet ownership and activity level:

Group	Highly Active	Not Highly Active
Dog owners	0.30	0.20
Cat owners	0.15	0.15
No pets	0.05	0.15

- Compute the probability that a randomly selected household is highly active.
- Compute the probability that a household owns a dog given that it is highly active.

Try this question on your own, and reach out if you have any questions!